Aryan Gautam

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Education

University of California Santa Barbara, BS in Computer Engineering Aug 2024 – May 2028

- GPA: 3.57/4.0
- **Coursework:** Computer Programming, Data Structures and Algorithms, Advanced Object-Oriented Programming, Computer Architecture, Comparison of Learning Algorithms, Computational Theory, Analog and Digital Circuits, Circuits Laboratory, Differential and Integral Calculus, Linear Algebra, Differential Equations, Classical Mechanics, Electricity and Magnetism, Waves and Optics

Technical Experience

Gaucho Rocket Project Aug 2026 – Present

Physics Department, University of California – Santa Barbara, CA

- Develop FPGA-based avionics systems for real-time rocket telemetry and flight data acquisition.
- Integrate STM32, ESP32, and Arduino microcontrollers with onboard sensors to collect and process real-time data from the rocket's avionics bay.
- Design and test embedded hardware and data-acquisition systems for reliable sensor monitoring during rocket operations.
- Work with real-time data streams and embedded systems to support flight instrumentation and avionics development.

Manufacturing Intern Jul 2026 – Sep 2026

Bhola Maharaj – Jamshedpur, Jharkhand, India

- Worked with a 95-year-old food manufacturing firm to standardize recipes and production processes for multiple varieties of sweets.
- Developed standardized formulations and process specifications to improve consistency, scalability, and quality across production batches.
- Explored and implemented automation solutions to produce different sweet products, improving manufacturing efficiency and reducing manual effort.
- Collaborated with production staff to evaluate workflows and identify opportunities for process optimization and automation.

AI Automation and Systems Intern Dec 2025 – Present

Synthesis AI – New York, NY

- Configure, test, and validate AI voice automation workflows to ensure accuracy, reliability, and real-world usability.
- Develop backend logic scripts and configuration pipelines, improving automation accuracy and reducing system errors.
- Use Git/GitHub to manage changes, create pull requests, and collaborate with engineers and business stakeholders to deliver customized AI voice solutions for 20+ clients, including testing, optimization, and backend scripting.
- Analyze performance metrics and behavior logs to identify bugs, trends, and improvement opportunities, documenting issues and communicating fixes clearly.

Research Assistant Oct 2025 – Present

Physics Department, University of California – Santa Barbara, CA

- Collaborating with the UCSB High Energy Physics group on the MilliQan experiment, which aims to detect millicharged particles using advanced scintillator and photomultiplier systems.
- Studying the effect of temperature on detector readings for muon particles to assess environmental influences on signal stability and detector calibration.
- Predicting an improvement of approximately 3% in the quality of data collected by the actual detector at CERN due to temperature optimization insights.

Making Data-Driven Decisions Jun 2025 – Sep 2025

MIT Schwarzman College of Computing - Massachusetts

- Developed a recommendation system to suggest optimal Amazon products to users, leveraging multiple techniques including ranking-based recommendations, similarity-based methods, and matrix factorization.
- Applied machine learning to segment customers, enabling personalized marketing, retention strategies, and optimized inventory and pricing.
- Built a machine learning model to predict high-potential leads, identifying key factors driving conversion and creating profiles of likely customers, achieving a 3% improvement in lead-to-customer conversion.

Electrical and Computer Engineering Project Jan 2025 – Mar 2025

ECE Department, University of California – Santa Barbara, CA

- Designed and built a remote-controlled boat, creating the main body using CAD software and integrating all electronic and mechanical components for full functionality.
- Was responsible for programming and writing the code for the Arduino and STM32 microcontrollers to enable wireless communication and real-time data exchange.
- Tested and optimized the controller–motor interface**,** enhancing transceiver communication reliability and extending the boat's operational time from 2 to 3 hours**.**

Research Experience

Research Publication Jun 2025 – Sep 2025

Department of Electrical and Computer Engineering, University of California – Santa Barbara, CA

- Proposed and designed a novel, hardware-inspired architecture for efficient on-device LLM inference, reducing redundant computations and improving real-time responsiveness by 2.7 percent.
- Developed a chatbot to facilitate communication between clients and users, leveraging the Blackberry Push system as a model and adapting its features to improve chatbot interaction with users**.**
- Optimized the chatbot system's memory efficiency by implementing cached memory on the main server**,** reducing redundant computation by 13% for repeated user prompts.
- Planning future enhancements to make the chatbot model scalable and to further validate its performance and capabilities**.**

Research Intern Jul 2025 – Aug 2025

UCSB Summer Research Program & KSE Global – Santa Barbara, CA

- Mentored 16 Mentored 16 international students, helping them adapt to the American education system and develop technical skills in machine learning and data science**.**
- Guided students in machine learning projects, teaching key concepts and equipping them with tools such as Python, pandas, scikit-learn, and matplotlib, while applying agile methodologies to analyze datasets, visualize patterns, and support evidence-based decision-making.
- Received "Best Intern" award for exceptional student engagement and for contributing to the Best Mentoring Section during the internship.

Research Publication May 2025 – Dec 2025

Department of Electrical and Computer Engineering, University of California – Santa Barbara, CA

- Conducted large-scale performance testing of AI models (ChatGPT with and without long-term memory) across 3,000 prompts, applying advanced algorithms and data structures to evaluate system efficiency.
- Developed Python-based pipelines (pandas, scikit-learn) in Google Colab to analyze model behavior and optimize inference workflows, aligning with embedded AI efficiency goals.
- Empirical analysis revealed that memory played a minor role in the actual response quality of the chatbot**,** providing insights into where system optimization efforts are most impactful.
- Developed Python-based pipelines (pandas, scikit-learn) in Google Colab to analyze model behavior and optimize inference workflows, aligning with embedded AI efficiency goals.

Research Publication Oct 2024 – Mar 2025

Department of Electrical and Computer Engineering, University of California – Santa Barbara, CA

- Identified a critical gap in student learning, finding that a procedural understanding of limits was insufficient for conceptual comprehension, particularly with asymptotic and non-real number concepts.
- Authored a research paper proposing a structured pedagogical path for professors, complete with examples and questions designed to foster a deeper, more intuitive grasp of limits.
- Cultivated essential research skills, including hypothesis formulation, data analysis, and professional academic writing, through the successful completion of this project.

Work & Leadership Experience

Resident Assistant Sep 2025 – Present

UCSB Housing, Dining & Auxiliary Enterprises, University of California – Santa Barbara, CA

- Foster a safe, inclusive, and engaging residential community by implementing programs, conflict resolution strategies, and student support systems.
- Serve as a primary resource for 76 residents, providing guidance on academic, personal, and interpersonal challenges while mediating conflicts and upholding university policies.
- Plan and execute educational and social events to enhance student engagement and community cohesion, developing leadership, problem-solving, and crisis management skills.

Association for Computing Machinery Jul 2025 – Present

- Serve as Research Branch Officer, mentoring 15 students and guiding them to actively contribute to university-level research projects.
- Responsible for organizing biweekly research symposiums and coordinating with professors to present their research to students, with a focus on engaging freshmen.
- Serve as Editor of the monthly newsletter, curating content and ensuring timely publication.

Institute of Electrical and Electronics Engineers Sep 2024 – Present

- Active member of IEEE on campus, contributing to the Tesla Coil project.
- Serve on the testing and troubleshooting team, performing oscillation frequency tuning and insulation verification to ensure safe and efficient high-voltage operation.

Publications

From Push Email to Push AI: BlackBerry-Inspired Efficiency in Large Language Models (International Journal of Engineering Research & Technology)	Sep 2025
Comparative Analysis of Chatbot Architectures with and without Memory (Zenodo)	Dec 2025
Why Asymptotic Behavior of Limits is Important for students to learn before Differentiation (International Journal of Research in Engineering & Science)	Jun 2025

Licenses & Certifications

International Applied Artificial Intelligence Institute Level 1 Certification	Sep 2025
Adult First Aid/CPR/AED {American Red Cross}	Jul 2025
Google Advanced Data Analytics Specialization	Oct 2023
AI For Business Specialization {University of Pennsylvania}	Sep 2023
AI in Healthcare Specialization {Stanford University}	Sep 2023
Google Data Analytics Specialization	Aug 2023
IBM Cybersecurity Analyst Specialization	Aug 2023